# DG INTERNS HUB — TECHNICAL NOTES

## TOPIC 08: THRESHOLD-BASED VS CORRELATION SEARCH ALERTS

**Submitted By:** Gunjan
**Internship:** DG Interns Hub Cybersecurity
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## 1. Detailed Comparison Matrix

| Evaluation Metric | Threshold-Based Alert | Correlation Search Alert |
|---|---|---|
| **Detection Mechanism** | Single event type count exceeding a fixed numeric limit | Multi-source log evaluation across defined temporal sequences |
| **Complexity** | Low (Simple single-source aggregate) | High (Complex relational logic, joins, lookups) |
| **False Positive Susceptibility** | High (Triggers on legitimate batch operations or typos) | Low (Requires multi-vector evidence before alerting) |
| **Best Used For** | Basic availability, brute-force attempts, spike checks | APT detection, Insider Threat, Ransomware execution |

## 2. Practical SOC Scenario: False Positive Reduction

**Scenario: Admin Password Reset vs Real Brute-Force Attack**

**Threshold Rule Alert:** Alerts whenever > 10 failed logins occur. A user forgetting their password triggers a False Positive alert for Tier-1 analysts.

**Correlation Rule Alert:** Requires `> 10 Failed Logins` AND `1 Successful Login` AND `Subsequent Admin Group Modification (Event ID 4728)` within 10 minutes. Ignores forgotten passwords and alerts only when unauthorized escalation occurs.

## 3. Recommendation for SOC Teams

Use threshold alerts as low-fidelity warning indicators, and leverage correlation searches for high-priority P1/P2 actionable incident tickets.